THE GRAMMAR OF EVOLUTION IN THE EVOLUTION OF GRAMMARS

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Abstract

Mental grammars are complex, self-replicating cognitive systems that exhibit characteristics typical of (cognitive) evolution. Principles and theorems of the theory of evolution explain the diachronic dynamics of grammatical systems. Especially the components of grammars that are subserved by the procedural memory system are shaped by evolutionary processes known from Darwinian evolution. Same problem structure, same solution structure. The aim of this article is to show that this is the case indeed.

Classical theorems of biological evolution and population genetics are presented as explanatory principles for the origin and development of grammars based on cognitive evolution: Fisher's fundamental theorem, Price's theorem, Wright's shifting balance theory, Baldwin effect, Dollo's law, Gause's law, Maynard-Smith's evolutionary stable strategies, Williston's law, Schmalhausen's principle.

1. Background

There are good reasons to assume that the grammars of natural languages are largely the result of Darwinian *cognitive* evolution (Haider 2015, 2021, 2023). Cognitive evolution is to cognitive systems what Darwinian evolution is to biological systems.[1] As for the latter, the principles of the Modern Synthesis provide the framework for understanding the processes of evolution. The Modern Synthesis of Darwinian evolution is the unification of Charles Darwin's theory with the discoveries of post-Mendelian and population genetics. This framework provides a comprehensive theory that explaines how variations arise and how evolutionary processes, particularly natural selection, drift, and flow act on these variants and – by chance and necessity – produce (adaptive) evolutionary changes over time.

Darwin's theory of evolution is substance-neutral, as he himself emphasised.[2] Modern biology agrees.[3] Variation plus natural selection, drift and flow shape *cognitive* systems according to the same systemic principles as they shape biological systems. Evolution is the theory of the dynamics of self-replicating systems, embedded in their respective selection environments. In this article, classical laws of the theory of evolution in the (extended) Modern Synthesis – cf. Wright 1931, Huxley 1942, Mayr & Provine 1998, Gould 2002, Pigliucci & Müller (eds.) (2010) – and several of its theorems will be confronted with the phenomenology of grammar change. If it is correct to assume that the processes of grammatical change are processes of

[1] At least since the 1990s, there has been speculation in the grammar-theory literature that grammatical change is a Darwinian evolutionary process at the *genetic* level. However, there is no trustworthy evidence to support these ideas. Grammatical changes do not leave genetic traces, and complex systems do not suddenly arise out of nowhere in a single species. Grammars do not evolve by *genetically-based* evolution.

[2] With regard to language, Darwin (1875: 61) made it clear that his theory of evolution based on natural selection is not bound to any particular substance: "*The formation of different languages and of distinct species, and the proofs that both have been developed through a gradual process, are curiously parallel*."

[3] "*Biologists will no doubt be thinking of numerous parallels between linguistic and biological evolution.*" Bowern (2015: R41). However, *cognitive* evolution is an autonomous domain of evolution; see Hunley et al. (2012).

evolution, then their empirical characteristics must correspond to the characteristics of evolutionary processes as defined by these principles of evolutionary theory.

Why would this be so? What would make grammars a target and product of (cognitive) evolution? Some of the background and answers have been presented already in previous articles that are readily[4] accessible. The following section provides background information to aid in understanding the subsequent discussion of the laws of evolutionary theory.

One point must be emphasised from the outset. We are talking here about evolution as a reality and not as a metaphor. Not every linguistic change is the result of evolution. Here, we are primarily concerned with grammatical change. These are changes in the process structure of grammars that occur without the users of the grammar being able to know what is going on. From a neurocognitive perspective, these are changes that take place in the consciously inaccessible, domain-specific capacities subserved by the procedural-memory network. However, consciously, and therefore intentionally, accessible components such as the lexicon are not excluded, since they also involve the procedural network, as in the case of storage and retrieval processes.

2. The ‘viral’ pathway to the brain – Grammars as mutualistic symbionts

Mental grammars are neuro-physiologically *embodied* (= ‘compiled’) as mental “language-apps”. These are *domain-specific* packages composed of, and running on, *domain-general* neuro-cognitive processing routines that enable, facilitate and perform language processing. As for their mode of replication, these packages behave like virus programmes. Viruses are complex, self-replicating information units that depend on a host to reproduce, and so are grammars. They spread from host to host, assisted by each host. Grammars, as virus programs, "infect" their hosts, but of course they do not do so in the same way as biological viruses. “Infection” means occupation with subsequent proliferation.

The way a *biological* virus operates is aggressive. It enters the host cell like a trickster and causes it to take over the virus's replication program. Mental grammars, on the other hand, are mutualistic symbionts, that is, *mutualistic cognitive* viruses entering into a symbiotic cooperation with the host. The hosts gain benefits. The mental-language app enables the brain to process verbal information effectively and efficiently. On the other side, the brains that uses the programs guarantee the replication of the programs by using it and producing ‘contagious’ input (= speech) for other brains. This works in vertical transmission (next generation) as well as horizontally (peer generation).

The cognitive-virus program is a program package that induces its hosts to compile it over a longer period of time as part of feedback-controlled processing of input, also known as language acquisition. The time window is tuned to an essential conservation mechanism (epigenetic pathways of myelinisation & synaptic pruning; see “swamping argument”, below). The trigger and driving force behind these processes is the children's tireless urge to interact verbally with their human environment as socially dependent members.

Therefore, “*we may need to concentrate less on the way in which we as a species have adapted to the task of using language and more at the ways in which languages adapt to being better*

[4] https://ling.auf.net/lingbuzz/_search?q=Haider

passed on by us." Kirby (2001: 110). "*Overall, language appears to have adapted to the human brain more so than the reverse*." Schoenemann (2012: 443).

Language acquisition is a process of highly, but not completely accurate replication of a grammar in the brain of a host. The inaccuracies give rise to a spectrum of variations in the speech community. The acquired grammar(s) then shape(s) the linguistic productions of their hosts, which constitutes the input environment for others acquiring language, and so on and so forth.

As the mathematicians Serva & Petroni (2008:1) emphasise, "*The evolution of languages closely resembles the evolution of haploid organisms.*" Viruses are haploid. Close structural parallels between viral evolution and language change are demonstrated in their study in which they apply a metric of computer science that quantifies the minimum number of single-symbol edit operations, known as Levenstein distance. They apply their method to the automatic creation of phylogenetic trees, in genetics as well as in diachronic linguistics.

Let me now address the crucial point: The two systems – biological organisms as well as cognitive systems – are completely different in terms of their material substance. The decisive commonality is the viral mode of replication. Since evolutionary theory is the theory of the origin and development of self-replicative systems, the same laws apply to both systems, that is, to biological viruses and cognitive ones. This will be demonstrated below.

One last reservation must not be missing here. Linguists who allude to evolution when discussing diachronic changes in grammars more often than not do not concern themselves with the principles of the theory of evolution. Their metaphorical language takes pleasure in the term "evolution" as shortcut for "development over time". Furthermore, guided by a functionalist view, they are led to confuse adaptation through natural selection with teleological change; for details, please consult Haider (2015:213-220).

Anyone who seriously deals with evolution proper outside of biology must first of all show how to avoid *Fleeming Jenkins' swamping argument*,[5] and second, abandon any functionalist presumptions. Linguists who speak of "cultural evolution"[6] are obviously unaware that the concept they use is a term without theoretical foundation. There is no evolution without *conservation* mechanism for a trait in the population. "Cultural evolution" does not have one.[7] It merely resembles the logic of variation, selection, and adaptation, but its way of spreading in a population does not involve replication and a conservation mechanism corresponding to inheritance.

In grammar change by *cognitive evolution*, the most important conserving force is the rigidity of the *procedural memory* system, which, in contrast to the declarative system, is consciously inaccessible. In L1-acquisition, grammatical routines are acquired and consolidated before the

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[5] The swamping argument is an objection raised against Darwin's theory by Fleeming Jenkin (1867), who asserted that an accidentally appearing profitable variety would not be able to spread in the population simply by natural selection. It would be 'swamped' in the pool of ordinary traits. Even if advantageous, it would be diluted each generation, until it disappeared again. What was lacking, as Darwin acknowledged, was a conservation device. In biology, this was identified only much later in the form of Mendelian genetic inheritance.

[6] Grammar as an achievement like pottering, stone throwing techniques, wheels, trains, frozen pizza, ...

[7] "*If the application of Darwinian thinking to understanding cultural change depended on the existence of replicators, we would be in trouble. Fortunately, culture need not be closely analogous to genes."* Boyd & Richerson 2000:158).

age around eight,[8] which marks the end of the period of myelinisation and synaptic pruning in the brain areas that embody the grammar processing network.[9] From then on, the grammatical routines prove to be highly resistant against changes. Changing them costs much devotion, time, and energy even to achieve only an approximation to the routines of an L1-acquired grammar.

As for functionalism, teleological or functionalist motivations are incompatible with the theory of evolution: "*Darwinian evolution is completely myopic*" (Simon 1996: 47).[10] Traits spread if the grammar variant that carries them has a higher replication rate than any other variant. First, their appearance is accidental (see: mutations). They can neither be planned nor staged. Mayr (1992) approvingly emphasises that Darwin explicitly disqualifies any appeal to function (viz. utility or final causes) in the quest for valid explanations: "*Nothing can be more hopeless than to attempt to explain the similarity of pattern in members of the same class, by utility or by the doctrine of final causes.*" Darwin (1875:191). If grammars are evolutionarily replicated, their diachronic development has no intrinsic direction. Change is a by-product of replication ecology, not an evolution toward ultimate perfection.

In the absence of variation, natural selection cannot operate, but variation cannot be arranged on demand. The highly misleading term "evolutionary pressure" is a common functionalist metaphor even in biology.[11] Properly understood, it means that the advantage a particular variant gives its bearer has *greater leverage* under certain circumstances, but such mutations can neither be planned nor induced.[12] If the environment changes negatively and beneficial mutations ('adaptations') don't happen spontaneously, such organisms will die out, no matter how intensive the alleged "pressure" is.

The functionality of grammars is a result of evolutionary adaptations to the selecting environment. This environment is the population of host brains of the grammar virus, that is, the brain routines recruited for language processing. Cognitive capacities that originally evolved for other purposes have been recruited for language, and grammaticalisation patterns as paths of evolution reflect the structure of these pre-existing systems. This is evolution by exaptation, which also applies to features that arise as non-adaptive by-products, aka spandrels; see Gould & Lewontin (1979), Haider (1991).

The history of the functionality of grammars is a history of how grammars adapt to the brain. Variants that don't cannot spread and are sieved out thereby. Dobzhansky's (1973) aphoristic title easily and appropriately translates into linguistics: Nothing in grammar theory makes sense except in the light of (cognitive) evolution.

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[8] "*The use of certain processing pathways during language acquisition is accompanied by epigenetic changes that consolidate these pathways."* Day & Sweatt (2011).

[9] This is, by the way, the approximate age limit at which a perfect, *spontaneous* acquisition of languages ends.

[10] "*Evolution depends on two processes: a generator and a test. The generator produces variety, new forms that have not existed previously, whereas the test culls out the generated forms so that only those that are well fitted to the environment will survive*." Simon (1996: 45). "*At each incremental step the evolving organism becomes fitter relative to its current environment, but there is no reason for the progress to lead to a global maximum of fitness of the individuals, separately or severally.*" Simon (1996: 47).

[11] "*FAULTY evolutionary explanations incorporating pressures and forces may be explained by the recruitment of everyday force-dynamic reasoning patterns into evolutionary explanations.* Nehm et al. (2010: 612).

[12] Friedrich Engels (1896) was a functionalist, of course, when he wrote: „*Das Bedürfnis schafft sich sein Organ*" ('The need creates its organ'). One and a half century later, quite a few linguists still don't know any better.

3. Principles of the theory of evolution

This section provides a short overview of basic principles of evolutionary theory in preparation for the following section, that deals with some of its theorems as the main section of this article under the maxim: Identical problem structure, identical solution structure. This is based on the idea that two superficially different areas have the same "solution structure" if they have the same “problem structure” and therefore the same underlying explanatory logic.

So, if a grammar is a complex, self-replicative entity whose phenotype is represented as the domain-specific cognitive capacity of any native speaker of a language with the given grammar, its explanatory modelling has a solution structure that is already pre-structured by the laws of Darwinian evolution and population genetics.

I.	Principles	Main mechanisms	Effects on evolution
1	Variation	Mutation, recombination	Creates diversity
2	Replication	Conservation	Maintains continuity
3	Mutation	Inaccurate replication	Introduces novelty
4	Drift and Flow	Sampling or migration	De-/In-crease of variation
5	Natural selection	Differential fitness	Adaptive change
6	Speciation	Isolation + selection	New species formation
7	Common descent	Inaccurate replication	Phylogenetic relations
8	Adaptive radiation	Rapid diversification	Multiple new species
9	Frequency dependence	Contextual fitness	Maintains diversity
10	Horizontal transfer	Transduction	Rapid spread of traits

Table I: Effects of Darwinian evolution – mechanisms & principles

3.1 Variation

Any field linguist is familiar with the fact that a "homogenous speech community" is just a linguistic idealisation. Real-world speech communities are always more or less heterogeneous. Boula de Mareüil et. al. (2021), for example, list twenty-four dialectal regions of Romance Italian dialects. A list of dialects of German[13] would cover several pages. Even the German standard variety comes in several variants. An educated Austrian, East Belgian, North German or Swiss person speaks the same language, but what they actually use are clearly distinguishable variants of the same ‘official’ language. Variation is the normal state, uniformity the exception. Variation affects all components of grammar, from phonology to morphology to syntax, and, of course, the lexicon. The diachronic development of grammars is divergent, rather than convergent. Likewise, random variation is an indispensable prerequisite for evolution. Without variation, there is no evolution by selection. In both domains, biology and linguistics, a principal source of variation is the not totally accurate transmission of information in its replication.

3.2 Replication

Evolution is a theory of the development of self-replicating systems. In biology, replication is most commonly achieved through genetic inheritance (sexual reproduction). However, this is

[13] https://de.wiktionary.org/wiki/Verzeichnis:Deutsch/Dialekte

not the only possibility. Viruses do not replicate through inheritance. They hijack a host cell's machinery to replicate its own genetic material and build new viral particles. The cognitive version of the viral way of replication is the way grammars replicate themselves as cognitive viruses. Grammars are the core component of the cognitive language "app" that every mature speaker possesses and uses after having successfully acquired it as a child. The linguistic productions are the input for the next cycles of L1-acquisition. Although ubiquitous, it is still not fully understood. How is it possible that a child with limited cognitive capacities can build up a complex capacity? Part of the answer comes from evolution: Only grammars that can be compiled already by a child's brain can replicate themselves. This has been the case since humans have had any language. Even the luxurious complexity of present-day grammars is an effect of evolution by natural selection and a reflex of the capacities of the human brain. Grammars are custom-fitted to the human language acquisition capacities by evolution. Trivially, unlearnable (traits of) grammars could not be learned and therefore not spread. In short, replication success is measured by the number of brains that the respective grammar variant is able to 'infect'.

3.3. Mutations

Mutation is one source of variation. Random changes in DNA sequences (mutations) are the ultimate source of all *genetic* variation. While most mutations are neutral or even harmful, some provide novel traits that may be advantageous in certain environments, fuelling evolutionary innovation. Grammar replication is accompanied by random changes in details of mental grammar programs that occur in L1 learners when they have to construct a replication of grammar using imperfect cognitive means. The results are reflected in the productions and can therefore find their way into other brains through their apparent reverse engineering attempts in the course of language acquisition, especially if they have positive effects on processing, storage, or retrieval. Another source of variation is flow, i.e. migration of features between dialects in mixed language communities.

3.4 Drift and Flow

In biology, *flow* (migration) is the horizontal transfer of genetic material from one population of a species to another. Gene flow can introduce new genetic variation into populations and counteract the effects of natural selection and genetic drift. The closest parallel in linguistics is the admixture and levelling of dialectal varieties or grammatical transfer in multilingual communities.

Drift is a phenomenon of evolution based on random fluctuations in the frequency of a particular variant (allele) in a population. The effects of genetic drift can be heavy, causing traits to become overwhelmingly frequent or to disappear from a population. This happens especially in bottleneck events, when a random sample of individuals either is (geographically) isolated or happens to be the only survivor. In linguistics, the effect of an evolutionary[14] drift can be seen in the comparison of isolated dialects with their 'mother' language.

[14] Please note that this concept of drift is different from E. Sapir's (1921) notion. A Sapirian drift is a vectored sequence of grammatical changes, as a consequence of variation plus natural selection; see Haider (2021:22).

3.5 Natural selection

Natural (= constant & blind) selection is Ch. Darwin's – and A.R. Wallace's – original idea and key concept, but it is easily misunderstood and therefore all too often misinterpreted. The idea behind it can be better expressed as "natural survival". It is not a targeted process of selecting certain favourable variants (as in breeding), but rather the accompanying and unplanned elimination of the least successful ones ('survival of the fittest' by elimination of the unfittest). Individuals with traits better suited to their environment tend to achieve more offspring. The differential reproduction/replication rate is the result of the mechanisms Darwin subsumed under natural selection, driving adaptive evolution.

For grammars, reproduction/replication means that some variants find more brains to host them than others and thereby spread, while other variants 'die out'. This is the viral way of survival. A successful virus infects more hosts than a less successful one, and so do grammars. Successful means better matching the demands of the selecting environment. The selecting environment for cognitive systems is the human brain, and in particular the computation routines suitable & recruited for language processing. Grammar variants that can be computed more effectively and efficiently and acquired more readily spread in the course of the evolution of a particular grammar.[15]

The grammar variants form a pool and inevitably compete with each other, simply because not all of them can find a host. This creates the conditions for selection, in which certain variants are preferred over others. The competition is as tough as for biological viruses. There is room for one variant of a grammar of a given language in the brain of a host. The winner takes it all. This variant will be conserved and it spreads if more brains host and use it, contributing to an input-environment for language acquisition with this variant (see also frequency dependent selection).

A common visualisation of the spread of a mutualistic symbiont through a host population *over time* is a sigmoidal curve (S-shaped growth curve). First there is a lag phase (slow start), followed by an exponential phase (rapid increase), that reaches a saturation phase (plateau), when the variant has infected nearly all host or filled all ecological niches. Any quantitatively well-studied grammar change displays this property, be it the spread of V-second in Middle High German, the development of *do*-support in Middle English – see Ellegård (1953:162), Haider (2021:24) – the rise of the English progressive ("*be* + V-*ing*"), to name just a few cases. Socially propagated changes behave differently because they involve threshold dynamics and frequency-dependent acceptance (people accept something because others do too; see fashion trends). Grammar evolution is driven by fitness differentials. A beneficial viral mutation doesn't become more beneficial just because it's common - whereas social innovations often become more attractive as they spread, a human consensus heuristic well-known as *bandwagon effect*.

3.6 Speciation

In evolutionary theory, speciation is the process by which populations become new and distinct species that are reproductively isolated, meaning they can no longer interbreed to produce fertile offspring. The linguistic counterpart to interbreeding is mutual comprehensibility of the

[15] Please note that in linguistics, "environment" is typically understood to mean the environmental factors of the respective language community, see e.g. Greenhill (2016), thereby neglecting (neuro-)cognitive aspects.

respective linguistic varieties. In this sense, Norwegian and Swedish are sub-varieties (i.e. subspecies), while Bavarian and Alemannic are different species, that is, different languages, even though they are listed as dialects of German. Swedish and Norwegian are mutually comprehensible, but Bavarians do not comprehend Alemanni, and vice versa, unless they are bilingual.

In linguistics, it is generally accepted that languages develop from out of a dialect continuum of a predecessor language. This has been confirmed beyond doubt by Indo-European studies. The present-day Indo-European languages can be traced back to a single proto language out of which the present-day languages ('species) have developed. A more immediately observable example is the development of Latin into various different languages (i.e. grammar 'species') after the end of the Roman empire with a sudden disruption of the ecologically stable situation of the Latin period (see Section 4.3).

3.7 Common descent and divergence

All living organisms share common ancestors, with species diverging through the accumulation of differences over time (see speciation). In Darwin's time, this was the counter-model to the steady-state theory, according to which the living world came into being in one fell swoop, through creation, and has existed in the same form ever since.

The idea that today's form is the result of a long history of more or less *gradual changes* is undisputed in linguistics, thanks to the successes of historical-comparative linguistics, in particular Indo-European studies. The creationist idea of an "Ursprache" (i.e. a unique proto-language or *lingua adamica*) was powerless against the empirical successes of comparative linguistics – until the Chomkyan turn, which gave us "Universal Grammar".

Suddenly, the creationist concept was back, disguised and hidden behind the how-else-arguments shielding the hypothesis of "Universal Grammar", as a *genetically* ingrained ability of a single species, that is, as "intelligent design".[16] No graduality, no descent, no evolution, just out of the blue, through a 'genetic' singularity. This marked the beginning of the estrangement between Generative linguistics and cognitive science.[17]

3.8. Adaptative radiation

Adaptive radiation is the rapid diversification of a single ancestral species into a multitude of new species that are each adapted to exploit different ecological niches. The grammatical counterpart to "ecological niche" is "neurocognitive processing mode." For example, if arguments are identified by structural configurations rather than by morphosyntactic markers (case, agreement), this corresponds to the use of different processing modes, or neurocognitive niches.

Eldridge & Gould (1976) developed the theory of punctuated equilibrium, according to which species remain relatively stable over long periods of time, interrupted ("punctuated") by short, rapid phases of significant change during speciation.[18] In essence, adaptive radiation provides

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[16] "*The arguments for* INTELLIGENT DESIGN [emphasis mine]HH *from irreducible complexity bear an uncomfortable similarity to that originally posited for the necessity of a genetically specified universal grammar.*" (Finlay 2009: 262).

[17] Although Generativists never tire of trying to ingratiate themselves with cognitive science, they are rightly ignored because they pursue a purely speculative approach, busily putting forward empirically unfounded, highly abstract theories from which arbitrary results can be derived.

[18] See O'Brien et al. (2024). Linguistic instances are presented by Atkinson et al. (2008).

a mechanism for the "punctuations" in punctuated equilibrium: the environmental shifts or new opportunities that drive sudden, clustered speciation after long periods of evolutionary stability.

What is predicted is that initial dispersal plus varying contact conditions consistently produce this radiation. Roughly, languages in high-contact zones (trade routes, multilingual areas) tend toward simplification and analyticity, while those in more isolated transmission chains maintain or elaborate morphological complexity. Consulting phylogenetic language trees provides a first impression, with the estimated time lines of the radiations of Austronesian, Bantu, Indo-European or Turkic languages. Predictions can be finally verified with phylogrammes in combination with chronogrammes (see Maurits et al 2020; Wilson et al. 2022). Currently available time series data are not meaningful for grammatical evolution, however, as they are mostly based on vocabulary comparisons, see Maurits & Griffiths (2014), but not on traits of grammars.

The split of the Germanic family into an OV and a VO group illustrates a feedback loop of change processes, as pointed out in Haider (2014): The spread of the V2 clause structure concealed the thereby abetted fixation of the previously flexible basic position of the verb, which opened up the option of V-initial vs. V-final. Both options found their replicatons.

3.9 Frequency-dependent selection

In the general selection process, the *fitness of a trait* is constant, no matter, how rare/common the trait is. Fitness is fixed relative to the environment. In frequency-based selection, the fitness of a trait changes depending on its frequency in the population. In this case, fitness depends – positively or negatively – on how common the trait is. It is important to understand the role of frequency in evolution.[19]

An obvious linguistic example is the replacement of "irregular" verb forms with forms from the regular paradigm. If it occurs rarely, it is replaced, but, if the irregular form occurs frequently in the environment, this stabilises its use. The causal mechanism is a trade-off between more 'costly' declarative storage and less costly just-in-time production according to a rule scheme. High frequency levels the storage and retrieval costs per item. So, it pays to store different form instead of a just-in-time production of forms. Typically, auxiliaries are irregular, or even suppletive verbs.[20] Another case is the singularity of the English verb "*believe*", which does not participate in the control construction (unlike *croire, credere, glauben, tänka*, ...).[21] In this case, schooling and literacy may be interfering factors, too. A final example is "*enough*" and its Germanic cognates. This is the only degree modifier that does not precede its target: big *enough* vs. *sufficiently* big. "*In all Germanic languages, the cognates of "enough" have survived and preserved their exceptional status over a period of more than a millennium, apparently due to its high frequency.*" Haider (2022: 199). Please note that the exceptional property is a grammatical property, namely the *word order restriction* of a single oddball item.

[19] It is not what linguists have in mind when they claim that there is a quantity-to-quality tipping-point effect in grammar change: A variant becomes more and more frequent until it is 'grammaticalised'.

[20] The 18th century physicist Georg Lichtenberg noted a correlation of frequency with irregularity: „*Diejenigen Verba, welche die Leute täglich im Munde führen, sind in allen Sprachen die irregulärsten* [Those verbs that people use daily are the most irregular in all languages]HH: *sum, sono, ειμι, ich bin, je suis, jag är, I am.“*

[21] "Donald believes [*himself* to be the greatest lover]" vs. "*Donald believes [*PRO* to be the greatest lover]"

3.10 Horizontal transfer

Horizontal transfer is recognised as a pervasive evolutionary process that distributes genes between divergent lineages. Horizontal gene transfer (HGT) is a specific type of gene flow. Gene flow typically occurs through vertical transfer (parent to offspring), but HGT happens through non-reproductive means like conjugation, transduction, or transformation.

The linguistic parallel is obvious, namely the grammatical interferences in multi-dialectal/multi-lingual communities. In a multilingual brain, there is more than one grammar package represented, and these packages influence each other during activations. Small and in some cases even larger units of one grammar can migrate into the other grammar(s). A telling example is the emergence of the V2 structure of German in Romansh, the Romance language of Grison in Switzerland; see Casalicchio & Kaiser (2023) due to the background of a multilingual speech community.

Horizontal transfer is a kind of shortcut. What is imported by horizontal transfer is itself an outcome of natural selection, but in another population. This shortens the path to a result that would otherwise only have been achievable through many intermediate steps, as Romansh shows. Importing the V2-grammar *as a package* saves the many steps that would be necessary for developing it.

4. Theorems of the theory of evolution explain grammatical changes

Let us recapitulate. For evolution to apply – understood as a general process of change in self-replicating systems under natural selection – certain minimal structural properties must be present. These are not limited to biology and can apply to any physical, informational, or artificial system that meets the following key requirements:

- **Replication** (self-reproduction): The system must have a way to produce copies of itself or its *informational structure*. Replication does not need to be perfect, but it must be reliable enough to maintain continuity over time.
- **Variation:** Replication *must not* be completely perfect; variations (mutations) or combinatorial changes must be able to occur. Variations provide diversity in replicated traits among replicators.
- **Conservation of transferred information:** There must be *a way to conserve information* (a "genotype" analogue) that is passed on to the replica during replication. This information must have influence on the structure or behaviour of the replicated system ("phenotype" analogue).
- **Differential fitness** (selection): Variants must differ in their ability to survive and reproduce in a given environment. The environment is selective so that some variants get favoured over others.

The target of the cognitive evolution of grammars is the mental-language app (and crucially not the speakers/listeners' environment). The selective environment for the "language app" is the neuro-cognitive ensemble of the human brain as the computational platform of the language app (rather than the communicative intentions or needs of speaker/listeners). Cognitive constraints shape which grammar types can successfully replicate, but successful grammars also shape which cognitive features provide advantages

The characteristics of evolution are fulfilled by the cognitively represented and functioning human language app, whose main component is grammar. We can therefore expect to find the typical characteristics of Darwinian evolution in the field of human languages, especially in the cognitively encapsulated areas of the procedural architecture of grammars. This expectation is answered in this section by examining theorems of evolutionary theory for their linguistic relevance and suitability.

4.1 Fisher's Fundamental Theorem

Let us start with the following question. In communities of which size do social innovations catch on faster, in smaller or larger ones? According to Trudgill (2002:103), when a population is relatively small, tight social networks can "*push through, enforce, and sustain linguistic changes which would have a much smaller chance of success in larger, more fluid communities*". However, 'linguistic change' does not refer to changes in grammar here, but to sources of *variation* in the speech community.

Sociologists tell[22] us also that small societies are faster in behavioural social diffusion, whereas large societies are slower, with more uneven changes. Diversity and scale would make change slower, because new habits may emerge in one subgroup but take a long time to spread to others, and institutions and traditions are more complex and resistant to rapid shifts. Linguistic examples of such socio-linguistic phenomena are easy to locate in the literature, cf. Labov (1990) or Svendsen & Jonsson eds. (2024).

How does all this relate to grammar change? If it were true that *grammatical change* is driven by socially conditioned desires regarding means of communication, grammatical change would essentially be a sociocultural phenomenon, with language as the main communicative tool. So, we would have to expect small communities to be grammatically far more innovative than large ones. If, however, grammar change is an instance of the Darwinian mode of cognitive evolution, the prediction is the exact opposite. This follows from Fisher's theorem:

(1) Fisher's Fundamental Theorem [Fisher (1930), Grafen (2020)]
$\Delta G/\Delta t = V(G)$ (G = genetic fitness, t = time, V = variance)

Interpretation: The rate of change of the mean fitness of any population is equal to the variance in genetic fitness. In grammatical terms this means that the rate of change of the mean neuro-cognitive fitness of any population of grammar variants is equal to the variance in the population of grammar variants.

Consequently, the more variation, the more chances for selection; the less variation, the less chances for selection. Therefore, the smaller the language community, the more resistant it is

[22] Regardless of theoretical background (diffusion-of-innovations theory, functional structuralism, social-capital theory, or symbolic interactionism) community size is a factor: A system with dense communication networks, like a "contagion" model of diffusion, can lead to faster adoption. Conversely, a system with low connectivity or limited communication channels will see slower diffusion; Dearing & Cox (2018).

to grammar change since there is little variation. What is the empirical situation? It clearly supports the evolutionary stance.

In every language family, there are languages that have changed much less than their cognate languages, and these languages are used by *small* communities. Here are examples of languages that have changed very slowly: Modern Icelandic and Faroese as Germanic languages; Logudurese Sardinian as a Romance language; Soqotri as an Arabic language; Lower Sorbian as a Slavonic languages; Gyalrong languages (85.000 speakers) as Sino-Tibetan languages.

Fisher's theorem correctly predicts such a situation. The more compact the language community, the more homogeneous its grammar stays over time. The smaller and isolated a language community, the better its language can serve as a window into earlier forms of the organisation of grammars. In small monolingual communities, variance is little. Therefore, the rate of change is proportionally little. Minimal variation means less chances for evolution. Earlier traits are retained, resulting in a smaller distance from the proto-language and giving the language a more archaic character overall.

4.2 Price's theorem (George R. Price)

Price's equation, in short, is a mathematical representation of "*survival of the fittest*". It simply says: Total change = change due to selection + change due to transmission. This factorisation helps us to understand the extent to which evolution occurs through natural selection, which favours certain traits, and/or through how these traits are passed on and modified between generations. For an application to language change see Gong et al. (2012).

(2) Price's equation (Price 1970, Frank 1995, Gardner 2008)
$\mathbf{\Delta z = Cov\ (w, z) + E\ (w\Delta z)}$[23]

Interpretation: This equation relates evolutionary change in a trait z to two components: *selection* (how the trait correlates with replicative success) and *transmission* (how the trait changes as it passes to offspring). The covariance term Cov captures natural selection favouring certain variants, while the expectation term E captures everything that happens during inheritance – mutation, development, and imperfect copying. "*The covariance and expectation in it decompose a dynamic process into a selection and an unfaithful replication component, and quantitative analyses of these components can lead to a better understanding of the effects of various factors on diffusion*." Gong et al. 2012: 11).

The equation doesn't care about the particular evolutionary mechanisms, it just partitions any observed change into these two fundamental components. This helps linguists move beyond just describing that change happened to understanding the forces driving it. The second component – unfaithful replication – is of decisive importance. Grammatical change is not brought about by natural selection through force, but only by invitation. The second component determines whether it takes hold and spreads.

A good illustration comes from co-existing, optional variants, as for instance, the order variants in verb clusters in Germanic OV-languages. Here, "survival of the fittest" is a neck-and-neck 'race' with several contending 'necks'. The causality is explicated in Haider (2010: 286-292;

[23] [Δz = change of a trait z; Cov (w, z) = covariance of fitness (w) of individuals and (z).
E ($w\Delta z$) = expected value of the product of (w) and (Δz)].

338-343). Here is just one illustration from Dutch (ANS 1997, vol. 2: 1069) next to German, in the right column (3):

(3)a.*dat hij niets [gezien hebben *kann*]	– dass er nichts [sagen müssen *wird*]
that he nothing seen have can$_{finite}$	that he nothing say must will$_{finite}$
'that he cannot have seen anything'	'that he won't have to say anything'
b. dat hij niets [gezien *kann* hebben]	– dass er nichts [sagen *wird* müssen]
c. dat hij niets [*kann* gezien hebben]	– dass er nichts [*wird* sagen müssen]
d. dat hij niets [*kann* hebben gezien]	– *dass er nichts [*wird* müssen sagen]

Both languages show variation in the order patterns of auxiliaries and quasi-auxiliaries in their positioning. The base variant in an OV language (such as Frisian, where it is the only variant) is (3a) and the predictable target is (3d). Dutch has reached it already and eliminated the base order (3a). (3d) is the complete reordering of the cluster structure, from left-branching into right-branching. German still uses the base order (3a), but in dialects (3d) can be found already. As long as (3a) does not give way, (3d) is out of reach for German.

What matters for the present concern is this. The gain in fitness, i.e. the bonus for right-branching clusters in processing and acquisition, is not strong enough for one of the variants to totally prevail. Gause's law (see below) has not the full grip on it yet. This follows from Prices's theorem, since for each of the competing traits z_{i-n}, the product $w\Delta z$ is too small to make an effective difference, except for the polar variants, that is (3a) in Dutch and (3d) in German.

4.3 Maynard Smith's (1974) ***Evolutionarily Stable Strategy*** (ESS)

An Evolutionarily Stable Strategy (ESS) is a strategy (or a set of strategies) that, if instantiated by a population, cannot be invaded by an alternative (mutant) strategy. Once an ESS is reached in a population, evolution comes to a local standstill, as no variant is available that could outperform an already established variant. This explains why evolution is not a perpetual motion machine. When a system has reached a stable equilibrium, it can remain in that state for a long time (cf. living 'fossiles' such as Latimeria chalumnae, existing since 409 million years).

In population biology, the concepts of ESS and an evolutionarily stable *state* are closely linked. Once virtually all members of the population follow this strategy, there is no 'better' alternative. In an *evolutionarily stable state*, a population's genetic composition is *restored by selection* after a disturbance, if the disturbance is not too severe, due to the available ESS. An evolutionarily stable state is a dynamic property of a population that returns to its previous stable state as an attractor.

What are ESSs in grammar theory? The major clause structure types, for instance, are evolutionarily stable states. They are not transitional. Another concrete example on a much smaller scale is English do support. It has turned in an evolutionary stable state during middle English within a few generations. Before, for instance in Shakespearean English, *do*-support was an alternative to the fronting of the finite verb in (yes-no-) questions. *Do*-support got grammaticalised and thereby, the position of the finite Verb was frozen in at the base position in the VP. As a consequence, English is the only Germanic language which has not become a V2 language.

When acquiring the first language (L1), the effect of ESSs determines the phases that children go through as they develop their way through the "adaptive landscape" (see the following section) to the target grammar. Even children acquiring an [S[VO]] grammar use SOV structures for some time as they are ESSs, as other languages show; cf. Hall et al. (2013), Langus & Nespor (2010), Schouwstra, & de Swart. (2014).

4.4 Wright's *Shifting Balance Theory* (SBT)

Wright's Shifting Balance Theory (SBT) explains how populations might reach, leave, or shift between ESS peaks due to drift, selection, and migration. The ESS theory and the SBT complement each other. The SBT is one of Sewall Wright's most influential contributions to evolutionary biology, and the *Shifting Balance Theorem* provides its mathematical foundation.

The theory specifically addresses what is described by Wright's famous "adaptive landscape" metaphor – where fitness is represented as peaks and valleys on a topographical map of genotype space. It proves that under certain conditions, populations can:

i. *drift away* from its adaptive peak, through random genetic drift in peripheral fractions of a population;
ii. *cross* an adaptive valley, where fitness is temporarily reduced;
iii. *climb* to a higher adaptive peak through natural selection from there.

The Shifting Balance Theory involves three phases. In *Phase 1*, random genetic drift in small, semi-isolated populations ("*demes*")[24] allows exploration of the adaptive landscape, potentially moving populations off local peaks. In *Phase 2*, selection drives populations that have crossed valleys up to higher adaptive peaks again. In *Phase 3*, the most successful populations spread their gene combinations to other demes through migration ("interdemic selection", that is, selection that acts on populations, i.e. demes, rather than on individuals). The theorem provides the mathematical justification primarily for phases 1 and 2, showing that the transition between peaks is theoretically possible. Here is a grammatical illustration.

Word order typology has identified three ESS optima ('peaks') for the positioning of phrasal heads, namely the phrase-*final* position (complement-head), the phrase-*initial* position (head-complement), and a *combination* of head-initial plus a single pre-head position (Spec-head-complement). They are stable but may nevertheless change, even in the absence of disruptive external events.

In VSO languages, a sub-population may introduce fronting items to the preverbal position of the V-initial clause (Phase 1), which allows exploration of the landscape and lead to a new adaptive peak, with the subject as the only or mandatory fronted item. The result is an SVO structure as the new ESS. Presently, most if not all VSO languages "*have SVO as an alternative or as the only alternative basic order*".[25]

In quite a few OV-languages, "extraposition" is an optional grammatical device for reducing centre-embedding. So, OV is partitioned into two subtypes. There is the rigid (strict) head-final type, including Japanese, Korean, or Tamil. On the other hand, there are non-rigid head-final languages, such as Avar, Basque, German, Persian, or Tsez. Presently, the type allowing extraposition (non-rigid V-final) is more frequent than the strictly verb-final type across the world's

[24] A *deme* is a local, partially isolated sub-population that can interbreed.
[25] Greenberg's 6th universal: "All *languages* with dominant *VSO order* have *SVO* as an *alternative* or as the only *alternative* basic *order*."

languages. Truly rigid verb-final languages that *never* allow any postverbal material are relatively rare. It is currently unclear whether "tightening" or "loosening" is the actual causal process. There is a lack of reliable empirical evidence; cf. Haider (2023: Section 1).

4.5 Schmalhausen's (1946) principle of stabilizing selection

Stabilizing selection favours the intermediate form of a trait over more extreme forms. It causes a reduction in the variation observed in the expression of this trait and enables the spread of the intermediate, better adapted trait in the population. Over time, intermediate forms become increasingly common among the population, while the extreme forms become rarer until they eventually disappear altogether. This is what we see in so-called standard languages in comparison to their dialectal spectrum, for instance.

Here is a fitting example from the dialect continuum of the Middle Bavarian dialects. In these dialects, the C°-position as well as its specifier can be simultaneously lexicalised in various ways. In standard German, doubly-filled Comps are ungrammatical; see Bayer (2025), Haider (2025). In general, for dialects that are grammatically close to the standard variety, the grammar of a construction in the standard language is the lowest common denominator of the corresponding dialectal constructions. Minority features that don't generalise well are no longer included in the standard version.

4.6 Dollo's Law

The principle was first stated by Dollo (1893) and later elaborated upon by Gould (1970). It states that complex evolutionary changes are effectively irreversible. Once a structure is lost, it doesn't re-evolve in the same form. "*An organism never returns exactly to a former state, even if it finds itself placed in conditions of existence identical to those in which it has previously lived [...] it always keeps some trace of the intermediate stages through which it has passed."* Gould (1970).

Here is a simple biological example. Mammals are descendants of fish, sharing a common ancestor with other vertebrates that evolved from a group of lobe-finned fish. When mammals reentered the water, they did not re-evolve as fish, although whales and dolphins superficially look like fish, for an obvious reason: Their lungs, which have evolved for air, are fundamentally different from the gill structures of fish, and re-developing them would be an extremely unlikely evolutionary path.

The development of gills on the one hand and lungs on the other is each an extremely complex trajectory through a system space for which a huge number of alternative trajectories are possible. So, for stochastic reasons, it is safe to assume that the age of the universe would not suffice as a time span for the evolution of a mirror image development of a previous series of complex developments. What we see here is a consequence of the 2nd law of thermodynamics.[26] Stochastically based processes spontaneously always move toward a state of *increased* entropy, that is, greater disorder.

A simple linguistic equivalent would be a language such as English, whose development reverses at some point and, after several millennia, achieves a grammar that is very similar or

[26] A principle that is known to be unknown in C.P. Snow's (1959) "other culture": $\Delta S_{univ} = \Delta S_{sys} + \Delta S_{surr} \geq 0$.

identical to that of Sanskrit. There is no step-by-step "rewind" to an exact restoration of a previous state.

So, Dollow's law is the principle that accounts for the *unidirectionality* of grammaticalisation. It is fascinating that exactly the same dogmatic debate about the universal validity of the principle is being conducted in evolutionary biology, e.g. Collin & Miglietta (2008) and in grammaticalisation research, e.g. Haspelmath (1999), Norde (2009). Evolution is no deterministic algorithm. It is a matter of "chance and necessity." However, there may be local steps that appear to be the inverse of steps in the developmental history of another species/grammar. Here is an example, viz. the diachrony of the French future tense, exemplified with the verb 'sing'.

(3) *cantābo* || cantare habeo[27]→ cantare aio → cantar ai → *chanterai* || je *vais chanter*

The Latin inflectional coding got replaced by a V + Aux combination. Aux-cliticisation led again to a *morphological* paradigm, which, today, is again in competition with an analytic form, that is, the (quasi-)auxiliary *aller* plus an infinitival verb. In scientific discussions, this has led to the interpretation that there is an iterative sequence of *synthetic* → analytic → *synthetic* → analytic.

So, if the step from synthetic to analytic were *unidirectional*, a subsequent *analytic → synthetic* step ought to be excluded. However, this is not what Dollo's law says. It merely says that there is no way from *chanterai* back to *cantābo*. What the law does *not* say is that there are no countervailing processes in the totality of all observable evolutionary processes. All it correctly says is that evolutionary processes are acyclical with respect to the phenotype.

This is true for grammars as well. The synthetic formation of the French future tense is not a symptom of a general shift from the analytical to the synthetic mode in French. In spoken French, passé composé (= analytic) has replaced passé simple (=synthetic), and the 'new' analytic future tense with *aller* + $V_{Inf,}$ confirms the synthetic-to-analytic drift.[28] It is a *general* drift, in Sapir's (1921) sense, which is unidirectional.

Here is another well-known example, namely "Jespersen's cycle" of negation. This "cycle" is no phenotypical cycle. What we see is a sequence of steps consisting of additive amplification, attenuation of the original particle and its subsequent loss. However, the cycle does not return to its original starting point, as a cycle would do. We see that a sequence of processes can repeat itself, but with different forms. From a diachronic perspective, this creates the impression of a spiral-shaped development. This is consistent with the principles of evolutionary theory and in particular does not affect Dollo's law.

4.7 Williston's law

Both Williston's and Dollo's laws describe evolutionary ratchets, that is, processes that tend to move in one direction and are difficult or impossible to reverse. Williston's law describes the *forward process of specialisation* and reduction, while Dollo's law explains why *reversing this process* is so difficult. Once structures are lost or fused together, following Williston's trend,

[27] Originally, this was a modal infinitival construction, like English "*I have to sing*" or German "*Ich habe zu singen.*"
[28] The way prescriptive grammar handles the situation is a good example of the *Competitive Exclusion Principle* (see sub-section 4.8). Each variant is admitted, but assigned to a different grammatical 'niche'.

Dollo's law suggests they cannot be simply "undone." Together, they characterise evolution as a path-dependent process where past changes constrain future possibilities – evolution tinkers with what exists rather than redesigning from scratch. Williston (1914: 22) describes a directional trend in evolution.

The parts in an organism tend toward reduction in number, with the fewer parts greatly specialised in function. Serial homologous structures tend to become reduced in number and more specialised in function over evolutionary time. The principle can be summarised as follows: *Numerical reduction + functional specialisation = increased complexity through differentiation*.

II. **Biological Evolution**	**Cognitive evolution** (via grammaticalisation)
Repeated structures become fewer	Multiple lexical forms reduce to fewer grammatical markers.
Structures become more specialised	Lexemes become specialised as grammatical forms.
Functional differentiation increases	Distinct grammatical roles emerge (e.g. tense, aspect, mood).
Number of independent units decreases	Fewer, more specialised grammatical forms

Table II: Biological vs. cognitive evolution

Table II lists correspondences between biological and cognitive evolution. The immediate grammatical correspondences are obvious. Grammaticalisation reflects Williston's law in that it involves the *reduction* in number and increase in *specialisation* of linguistic elements. A locus classicus is the cross-linguistic patterns of grammaticalisation of auxiliaries for TAM systems. Typically, tense auxiliaries are recruited from verbs of motion and aspectual auxiliaries from verbs of possession or getting into possession.

At the outset, there are just multiple lexical verbs with broad meanings, such as motion verbs (go, come, walk, run, depart, arrive, enter, exit) or possession verbs (have, hold, get, take, possess, own), each verb as full lexical item with its own semantics. The next stage is specialisation and reduction. Specific verbs (typically "go," "come," "have") are recruited as quasi-auxiliaries for highly specialised grammatical functions (future, perfect, passive). The inventory of forms is reduced, but functional distinctions multiply. The resulting state is a differentiated grammatical system with a small set of auxiliary forms, precisely differentiated temporal, aspectual, and modal distinctions, and a greater systemic complexity despite (or through) formal reduction:

This satisfies Williston's Law: reduction in the number of independent lexical forms that can express temporal or aspectual notions, combined with an increase in specialised grammatical functions. The selection context is the shifting of declaratively coded information to procedural coding, which consists of a *single* exponent and a *rule* of combination.

4.8 Competitive exclusion principle (CEP)

This principle[29] states that species competing for identical resources cannot stably coexist. One will outcompete the other(s) – the winner takes it all – or they'll partition into niches, that is,

[29] Originally formulated by Grinnell (1904): "*Two species of approximately the same food habits are not likely to remain long evenly balanced in numbers in the same region. One will crowd out the other*". Nevertheless, it has been associated with the name of Georgii Frantsevich Gause, as Gause's Law.

use resources differently. In human prehistory, Neanderthals (Stringer & Crété 2022) and Denisovans (Hassett 2025) lost out in competition with Homo sapiens, who prevailed in the struggle for resources. In the history of grammar, a good illustration of this principle is the diachrony of future tense formation in the Romance languages.

Classical Latin had an inflectional ('synthetic') future tense. In Late Latin, competing periphrastic variants emerge. So, several periphrastic constructions (with various quasi-auxiliaries + infinitive) competed with the synthetic future for "the same resource", that is, being the grammatical marker of future tense: *habere, velle, vadere/ire, debere*, etc. Intriguingly, *habere* + *infinitive* won the competition in most of Western Romance while *velle* + infinitive prevailed in Eastern Romance (Romanian):

(4) a. Catalan: amaré < amare habeō
b. French: j'aimerai < amare habeō
c. Italian: amerò < amare habeō
d. Portuguese: amarei < amare habeō
e. Spanish: amaré < amare habeō
f. Romanian : voi ama < volō amāre

The *CEP dynamics* is as follows: During Late Latin and early Romance (roughly 3rd-8th centuries CE), these constructions genuinely competed for the same grammatical territory. Speakers had variant ways to express future. Over time, in the course of geographic separation (Romanian vs. Western Romance), one form typically excluded others. This is *competitive exclusion* operating across time and space in grammar, with the twist that the outcome varies by linguistic community – a case of *allopatric speciation*[30] in grammatical forms.

An intriguing case of niche formation is *split ergativity*. There are ergative languages in which non-pronominal arguments are case-marked in the absolutive-ergative mode, while pronominal arguments are coded in the nominative-accusative mode. The two modes co-exist in the same grammar, but evidently in different niches.

The CEP is challenged by rare exceptions. In biology, Hutchinson (1961) coined "*the paradox of the plankton*", which invited many alternative explanations; see Record et al. (2014). Numerous phytoplankton species coexist in seemingly homogeneous water, all competing for the same nutrient matter and light. A linguistic example of an apparent paradox is the rich morphological case inventory of Icelandic, with a number of ('quirky-case') verbs, whose highest-ranking argument in the argument-structure is Accusative or Dative. But, even the dense communication networks on a small island cannot prevent a 'sickness' from spreading among the younger generation; see Eyþórsson (2001), on Dative Sickness (*þágufallsveiki*), as the tendency for dative case to replace lexically specified accusative case in certain constructions, and Nominative Sickness (*nefnifallsveiki*) as the tendency for non-nominative subjects in impersonal constructions (originally dative or accusative) to become nominative.

In the first tape, the dative wins as the standard case for a lexically case-marked argument since the accusative is the typical structural case. In the second type, the nominative as subject case wins over the dative in the subject function. Ultimately, as predicted by Gause's law, the

[30] Allopatric speciation is the process where a new species evolves due to geographic isolation from its ancestor

nominative will once be the only subject case (except for some highly frequent verbs perhaps; s. frequency selection). Today's Icelandic is a window onto a gradual change in case management that is currently taking place.

4.9 Baldwin Effect

The interaction between learning and *genetic* evolution is a central aspect of evolutionary theory and therefore also of the theory of *cognitive* evolution, as it deals with how the behaviour of the phenotypes can influence the genotype. The Baldwin effect describes how learned behaviours can influence genetic evolution by creating a selection environment that creates a bonus from which individuals benefit who can acquire those behaviours more easily. In the standard Baldwin effect, learned behaviours that improve fitness eventually become genetically assimilated because individuals with genetic predispositions toward those behaviours spend less energy acquiring them. For *viral grammars*, the corresponding process works as follows:

1. Any (proto-)grammar initially required a high degree of cognitive effort in learning and using (phenotypic plasticity).
2. Grammars that can exploit learnable cognitive strategies replicate successfully.
3. Over time, grammars itself evolve to be more "learnable" – incorporating features that align with cognitive biases or shortcuts that hosts naturally employ.
4. Eventually, grammars become so well-tuned to exploit existing cognitive machinery that acquisition appears nearly automatic.

In their evolution, grammars exploit whatever pattern recognition, statistical learning, or cognitive shortcuts are available in host brains. If children naturally segment speech into units based on prosodic cues, grammars would evolve to align syntactic boundaries with prosodic ones. If humans have domain-general sequence learning abilities, grammars will evolve structures that fit these patterns, perhaps explaining why dependency lengths tend to be minimised cross-linguistically. Any cognitive "free ride" – operations the brain performs effortlessly – will be exploited by successful grammars.

The Baldwin effect predicts grammars to become progressively more efficient at replication. Complex morphological systems (due to phonological ‘deterioration’), for example, evolve toward configurations that can be acquired with less explicit learning, through increased regularity. This explains why analytic (= procedural) structures replace synthetic (= declarative) ones in language history. Morphological case systems (= declarative) get replaced by structurally coded grammatical functions (= procedural).

Grammars do not have their own cognitive machinery. They evolve to parasitise whatever processing routines are available: Event sequencing abilities are exploited for tense/aspect systems, social cognition (theory of mind) is repurposed for evidentiality and perspective-marking, chunking is the information processing technique that leads to phrase-structuring, and so on. This matches observations that grammatical systems recycle cognitive resources evolved for other purposes. Languages are parasitic. There is no “language organ”.

Since all grammars face the same “host environment” (i.e. the innate human cognitive architecture), Baldwin-effect selection will drive them toward similar solutions. This explains linguistic universals and statistical tendencies across unrelated language families – not because they’re innately specified, but because they represent *convergent evolution* toward optimal exploitation of human cognition. Different language families discovering similar grammatical solutions reflect different "viruses" independently evolving to exploit the same cognitive affordances

At this point, a potentially unwelcome[31] prediction should not be omitted. If this process has been operating over the ~300,000+ years humans have had complex language, grammars should have become progressively better at replicating themselves. Modern grammars are predicted to be better "tuned" to human cognition through millennia of selection than grammar of languages spoken very many generations ago.

The Baldwin effect also explains one of the most puzzling features of human language, namely, why children acquire overly[32] complex grammatical systems with apparent ease and minimal explicit instruction. A momentarily widely accepted answer invokes an innate Universal Grammar, but the viral Baldwin effect offers a better alternative: It's not that children's brains are pre-loaded with a grammar blueprint. Rather, grammars have evolved over many generations to be exquisitely learnable by exploiting whatever learning mechanisms children do have. The "poverty of stimulus" problem dissolves if the grammar itself has evolved to be acquirable from exactly the kind of impoverished data children receive.

Within a single grammar, there is a Baldwin-effect selection on individual features: Grammatical constructions that are easier to acquire are used by more speakers and transmitted more faithfully. Complex constructions requiring extensive learning persist if they provide sufficient communicative value, but selection will simplify or regularise them. This creates frequency-dependent selection within grammars: common patterns become easier to learn because the grammar evolves to make them easier, which makes them more common (a feedback loop).

There is a Baldwin effect operating on *both* sides simultaneously. On the *host side*, human cognitive architecture is predicted to evolve genetic adaptations that make language learning easier (the traditional Baldwin effect for a language faculty); on the *virus side*, grammars evolve to exploit whatever cognitive mechanisms exist in hosts (viral Baldwin effect).

These two processes create a co-evolutionary mutualism: As human brains evolve better language-learning mechanisms, grammars can become more complex while remaining learnable. As grammars evolve to be more exploitative of cognitive shortcuts, there's less selection leverage ('pressure') on humans to evolve specialised language mechanisms. Eventually an equilibrium is reached where grammars are optimally tuned to human cognition, and human cognition is optimised for hosting these particular types of grammars.

Crucially, the viral-side Baldwin effect operates much faster than the host-side Baldwin effect: Human genetic evolution takes thousands of generations. Grammar evolution can happen in dozens or even single generations. This means grammars can "adapt to" human cognition much faster than humans can adapt to grammar. The system will be dominated by viral adaptation, with human cognitive evolution providing only a slowly-changing background constraint

[31] Some claim dogmatically that all human languages are equally complex. Even if this were the case, it does not mean that they are equally easy to acquire or equally efficient in encoding information.
[32] Grammars are more complex than communicative needs would require. This parallels, for example, the luxurious characteristics of coral reef communities. Evolution exploits the potential of a system, without self-interest.

Eventually, we may ask ourselves *why are languages that diverse?* If grammars evolve under Baldwin selection to optimise learnability, why do we still see such vast typological diversity? The answer is "evolution". For one and the same optimisation challenge, there is always more than one solution (multiple peaks in the adaptive landscape). Evolution is not deterministic. An additional source is historical contingency and path dependence: once a grammar goes down one evolutionary path, switching to another equally optimal path might require passing through less-learnable intermediate stages. Different populations may have subtle cognitive differences during childhood (from culture, environment, or genetics) creating slightly different selection landscapes. Finally, some variations are selectively neutral from the point of view of learning ability, and their spread is a matter of chance.

5. Mutualistic-virus model vs. functionalist tool model – A comparative assessment[33]

Now that sufficient evidence has been presented for a theory on the origin and change of grammars within the framework of a substance-neutral Darwinian evolution, it is time to systematically analyse its explanatory power in comparison to the widely entertained notion of language shaped by its users as a socially designed tool. The opposing viewpoints are obvious.

Here – the virus model (VM) – is a hypothesis of the cognitive evolution of grammars, embedded in the theoretically precisely formulated and empirically well-founded framework of evolution. *There*, is a common sense-based functionalist conception of language, upheld as a communication tool that people shape according to their needs, in the "form-follows-function" creed. Grammars (will) meet the needs and preferences of their users.

The two positions are so far apart that it should be easy to determine which of the two is fundamentally more consistent with linguistic reality. It will come as no surprise that this is the theory of the cognitive evolution of grammars as symbiotic cognitive viruses. Here is a summary of the parameters of comparison to be discussed.

III.	Phenomena	**Tool model** (TM)	**Virus model** (VM)	Better
1.	Maladaptive complexity	Should be eliminated.	Can persist if replicable.	VM
2.	Innovation resistance	neophile	neophobic	VM
3.	Child vs. adult role	adults primary	children primary	VM
4.	Structural correlations	modular	co-adapted	VM
5.	Optimal complexity	communicative efficiency	transmission bottleneck	VM
6.	Diversity patterns	functional	reproductive isolation	VM
7.	Individual variation	noise	quasi-species	tie
8.	Redundancy	processing aid	transmission fidelity	tie
9.	Speed of change	gradual	punctuated	VM
10.	Universals	optimal solutions	host susceptibility	VM

Table III: Tool model (TM) vs. mutualistic-virus model (VM)

[33] The author acknowledges the use of AI systems (Claude, ChatGPT, Gemini) for fact finding in this comparison. The final content was created, edited, and verified solely by the author.

5.1 Maladaptive complexity

VM prediction: Since selection acts on replicability/transmissibility rather than utility, grammatical features can persist despite being communicatively useless, or even dysfunctional and burdensome.
TM prediction: All complexity should serve communicative function (or be historical residue being eliminated). Useless features face constant pressure toward elimination.

Empirical reality: English is the only Germanic language in which intransitive verbs cannot be passivised, as a consequence of a grammatical glitch. A comparison with its closest related languages shows that English had run out of expletives for the subject position. Neither "*there*" nor "*it*" were available anymore; see Haider (2019). Another example is the coexistence of rich case morphology in Icelandic and Faroese with the [S[VO]] architecture that codes grammatical relations structurally. It has become grammatical ballast, as the continental Germanic OV-languages show, but it is retained.
Better: VM explains why grammars retain costly features that would be "bugs" if grammars were engineered tools. If there is no suitable variant, there is no change; see Fisher's theorem.

5.2 Transmission fidelity vs. innovation

VM prediction: High-fidelity replication is fitness-critical for the given grammar variant. "Conservative" grammars (resistant to change) are favoured by evolution. Innovation is dangerous to viral fitness, since most mutations are harmful. So, mechanisms that resist change are expected.
TM prediction: Innovations should be welcomed as better tools in cultural evolution. Change driven by communicative needs should be rapid. Mechanisms that facilitate adaptive change are expected.

Empirical reality: Most innovations are rejected. Successful *grammatical* changes are rare. Speakers act as "immune system" defending their grammar. Languages resist borrowing grammatical structure far more than lexical items. Stasis is the rule, not the exception.
Better: The virus model explains why grammatical change is rare and takes several generations. The conservatism of grammar makes sense for self-replicating information, less so for tools that should improve readily, within one generation, as changes in vocabulary show.

5.3 Child vs. adult role

VM prediction: *Children* are the *primary* replication machinery in vertical transmission and horizontal transmission (peer effects). Child acquisition patterns shape grammar evolution more than adult usage. The critical period is a narrow window of successful "infection". Adults are secondary hosts (more horizontal transmission by peers).
TM prediction: *Adults* are the *primary* users and innovators. Adult communicative needs should drive evolution. Children simply take over the tool adults have developed. No reason for critical period asymmetry.

Empirical reality: VM explains *critical period effects.* Only child learners achieve native grammar. Adults , on the other hand, may switch languages but impose substrate grammar; children acquire target grammar. Children's "errors" (= regularisations) often become the stable form in the next generation(s).

Better: Virus model explains why children are the crucial transmission bottleneck. This makes sense if grammar is a replicator, less so if it's a tool.

5.4 Structural correlations (Typological constraints)

VM prediction: Grammatical features *cluster* in co-adapted or sequence-adapted complexes. You can't mix-and-match freely (viral genome integrity). Cross-linguistic correlations reflect internal grammatical logic, not communicative optimisation. Example: Head-directionality parameters. VO languages vs. OV languages show correlated differences in adposition order, genitive order, etc.

TM prediction: Features should be *modular* - borrow the best available tool for each job. Correlations should reflect functional pressures (communicative efficiency). Languages should easily adopt useful features piecemeal.

Empirical reality: There are strong correlations in the grammatical architecture (cf. Greenberg's universals). Grammatical features cluster non-randomly. There is resistance to piecemeal grammatical borrowing. Languages readily borrow words but rarely borrow grammatical structure. Grammatical changes tend to affect multiple linked subsystems (cascade effects lead to sequential adaptations).
Better: Virus model explains this as co-adapted complexes, which suggests genome-like integrity, not tool modularity.

5.5 Optimal complexity level

VM prediction: Complexity is optimised for *learnability by children* (transmission bottleneck), but some "junk complexity" (parasitic features) can persist, too. No strong pressure toward minimal complexity. Equilibrium between mutation/drift adding complexity and acquisition bottleneck removing it.

TM prediction: Complexity is optimised for *adult communicative efficiency*, with a clear pressure toward communicative optimality. Redundancy is tolerated where it is functional (error correction, emphasis). Therefore, we should see active streamlining by adult usage.

Empirical reality: enormous variation in morphological complexity, from isolating via inflectional to polysynthetic systems is a fact, with no correlation between grammatical complexity and communicative success of speech communities. Icelandic (complex morphosyntax) and Mandarin (simple or no morphosyntax) function both perfectly well. Creoles show how morphologically poor grammars arise when the child-acquisition bottleneck is maximally constraining.
Better: The VM explains how complexity reflects replication ecology, not communicative optimisation.

5.6 Linguistic diversity pattern

VM prediction: Diversity arises from reproductive isolation (geographic/social barriers). Like biological speciation, there is allopatric divergence. Each "viral strain" optimises for its transmission ecology. Therefore, diversity is high for isolated populations (Pacific, New Guinea). Lower diversity is found where there has been recent expansion/mixing (colonial languages).

TM prediction: Diversity arises from different communicative needs. Languages should converge on optimal solutions. Contact should produce hybrid vitality (best features from each). Diversity reflects functional specialisation.

Empirical reality: Geographic isolation strongly correlates with linguistic diversity (New Guinea: 800^{+} languages). Language families show tree-like divergence (like viral phylogenies). Contact situations show more simplification than optimisation (pidgins/creoles). Convergence occurs but is superficial (borrowing features without deep integration).

Better: The VM explains why distribution patterns match reproductive isolation of replicators.

5.7 Individual variation and idiolects

VM prediction: Individual brains are hosts with slightly different "infections". Variation is normal (quasi-species cloud around consensus grammar). Stabilizing selection maintains coherence despite variation. Individuals may carry "latent" grammatical variants not fully expressed. TM prediction: Individuals should converge on community standard (tool-sharing). Variation is noise/error, not meaningful. Strong social pressure toward uniformity. Idiolects are just imperfect approximations of the community tool.

Empirical reality: Idiolectal variation is found even in linguistically stable communities. Individuals show variable grammar (sometimes *eclectically* accepting a novel structure, sometimes not). Covert prestige makes speakers maintain non-standard forms despite knowing standard. Individuals can harbour multiple competing grammatical systems (bidialectalism). Social pressure does not focus on grammar, but on social tasks, such as gendering, where grammatical violations[34] are accepted as a necessary evil.

Better: VM is slightly favoured. Treating individuals as separate hosts with variant-grammar infections explains grammatical variation naturalistically.

5.8 Function of redundancy

VM prediction: Redundancy aids error-correction in *transmission* (vertical transmission bottleneck). Multiple marking of same feature guarantees robustness against mutation. Agreement systems are an example of error-checking for viral genome integrity. The system becomes optimised for the ‘noisy’ channel of child acquisition.

TM prediction: Redundancy aids error-correction in *communication* (noisy speech signal). Multiple marking ensures that message gets through. Agreement systems help the parser to assign structure. The system becomes optimised for real-time processing

Empirical reality: Agreement often doesn't aid parsing because of neutralised forms and unpredictable gender categorisations. Redundancy patterns don't always match communicative frequency/importance. Some redundancy appears after a category is well-established (not for bootstrapping). Creoles eliminate much redundancy but communication works fine.

Better: VM is slightly favoured. Both models explain aspects of redundancies, but the virus model accounts better for non-functional redundancy and the existence of even formally

[34] Rule 9.4b of the R&A Rules of Golf: “*If the player lifts or deliberately* touches *their ball at rest …”* Here, “their” refers only to “the player,” and this is no typo but a politically correct (and grammatically wrong) way of avoiding *his*. In the golf text, only the player is meant, in contrast to the rules (of anaphora) of English grammar.

luxurious systems without communicative benefits. The VM explains the presence of redundancies through the contributions of different subsystems. One is subserved by the *declarative* memory network (morphology), the other by the *procedural* network (phrase and sentence structure). The redundancies result from an overlap in the operational areas. The TM has no answer for a simple question: Why aren't all languages like Mandarin Chinese, without any case and agreement morphology?

5.9 Speed of change

VM prediction: Change should show a punctuated-equilibrium pattern Stable periods (strong stabilizing selection) precede rapid change under stress (variation released, new equilibrium found). The rate is determined by transmission ecology, not communicative pressure.

TM prediction: Change should track communicative needs with continuous gradual improvements. Rapid change is predicted when technology/society creates new communicative demands. The rate determined by "functional pressure."

Empirical reality: Rapid change follows long periods of stability, see Old English to Middle English to Early Modern English. There is a contact-induced acceleration, with replicative stress that releases variation. In the Germanic area, the split into the OV group and the VO group (Dutch, Frisian, German; Afrikaans) went parallel to the rise of the V2-property. Evidently, neither OV nor VO, nor V2 (see English as the odd ball) are (dis-)favoured by their communicative effects. Cross-linguistically, V2 it is a rare grammatical phenomenon.

Better: VM predicts punctuated-equilibrium changes and fits Schmalhausen's stress-variation framework.[35] TM predicts socially conditioned patterns of changes, but there are no social correlates of VO, OV, or V2 (or not, as in English).

5.10 Cross-linguistic "attractors" = grammar universals

VM prediction: Grammatical universals are data structures that are optimally suited to the neuro-cognitive processing capacities. Consequently, cross-linguistically recurring patterns are successful viral strategies for replication in human hosts. Some grammatical "designs" are just better at spreading since they enhance infectability rather than communicative functionality.

TM prediction: Universal Grammar is the optimal *engineering* solution to *communication* problems. Recurring patterns are the convergent decisions on best tools. Languages independently discover good solutions. Constraints are about communicative functions.

Empirical reality: Some universals seem functionally arbitrary, i.e. with no clear communicative advantage, with arbitrary but stable correlations (head-directionality clustering). Rare typological patterns aren't obviously worse communicatively. Importantly, children appear to be biased toward certain structures before having developed their communication needs.

Better: Virus model explains the arbitrary but stable patterns as the result of host-parasite coevolution rather than communicative-functional optimisation.

[35] The stress variation model suggests that environmental stress can promote new phenotypic traits that increase variation. This process involves an initial period of stress-induced phenotypic plasticity, ultimately leading to genetic assimilation. See Badyaev (2005), Potticary et al. (2020), Nijhout et al. (2021).

5.11 "Universal Grammar" is dispensable

Eventually, a remark is due to the "elephant in the room." The viral-grammar model offers a radically different explanation for Generativist assumptions about Universal Grammar. UG principles, i.e. cross-linguistic grammatical invariants, are not the reflexes of *innate* constraints on possible grammars. Instead, they are universals emerging from Baldwin-effect selection features that every successful "grammar virus" eventually evolves because they are much easier to acquire in this form. This is *convergent evolution*. The apparent "innate knowledge" children bring to language learning is actually the result of grammars having evolved to be acquirable using only domain-general learning mechanisms.

This turns the innateness argument on its head: grammar looks learnable not because brains are pre-programmed for it, but because grammar has been programmed (through selection) for brains. The viral Baldwin effect thus provides a potential mechanism for how grammatical systems can become progressively better at exploiting their hosts, leading to the seemingly miraculous fit between human cognitive capacities and grammatical complexity. It suggests that much of what looks like innate linguistic knowledge might actually be the accumulated result of countless generations of grammars evolving to be learnable by generic primate cognition – a form of "linguistic niche construction" where the parasite shapes its own selective environment.

Overall, the predictions from the UG perspective are contrary to the cognitive-virus perspective and most likely wrong. If UG were the task master of human grammars, they should *converge* on the default values. What we see is ongoing diversity, cf. Hammarström (2016), and this is predicted by the evolutionary environment.

5.12 Overall balance

The overall assessment is straightforward. The viral model explains a lot more empirical facts than the tool model (or the UG-model, as an instance of intelligent design), and does so in a way that can be understood and in principle be measured in detail. In particular, the aspects of grammar that appear communicatively "irrational" at first glance are decisive evidence for the virus model if they are advantageous for replication (maladaptive complexity, conservatism, acquisition-bottleneck centrality).

The tool model isn't totally wrong – languages DO serve communication and show adaptive design – but it is a secondary constraint on *primarily replicator-driven evolution* of grammars. They must be "good enough" tools to keep hosts using them, but within that fairly broad-spectrum condition, viral fitness dominates, that is, the ability to spread to more brains than other variants. This is precisely analogous to mitochondria: they serve the cell (energy production), but they're still semi-autonomous replicators with their own evolutionary interests.

5.13 Some empirically testable predictions

- Grammatical features that align with well-documented cognitive biases should be more stable and widespread.
- Neuro- and psycholinguistic correlates of lighter/heavier processing load should be used for experiments comparing data structures corresponding to those before and after a grammatical change.

- Historical language data should show grammatical changes tending toward increased regularity and cognitive alignment over time.
- Artificial language learning experiments should show that "errors" learners make systematically push languages toward more learnable configurations – exactly the viral Baldwin effect in action.
- Creoles and sign languages emerging de novo should rapidly evolve features that exploit cognitive biases, even if these weren't present in the input languages.

6. Concluding remarks

The famously futile search for a western sea route to India led to the discovery of a new continent; the futile search for an innate universal grammar led to a great deal of wasted time. Since the late 1950s, the appropriate theory – the Modern Synthesis – was already in full bloom, even at MIT,[36] but hardly any linguists took it up. Modelling grammars in analogy to computer programming was a stronger attraction (cf. sequential Von-Neumann computing as model for Generative Grammar). Those who tried to seek a connection to the theory of evolution failed because they applied concepts of *genetic inheritance* to the development of grammars, which is not the road to success. Why would someone who is genetically gifted at mastering verbal expressions with recursive embeddings raise more children than someone who does not have this gift? Neither grammars nor a supposed universal grammar are inherited.

Darwin's theory, generalised and transposed to *cognitive evolution*, is the key to a cross-disciplinary understanding of human language abilities. It enables the meaningful combination of findings from evolutionary biology and population genetics with those from experimental psycholinguistics and neurolinguistics, language typology, synchronic and diachronic linguistics, and grammar theory, to describe and theoretically model the grammatical systems of human languages. In other words, it is suitable for leading to a comprehensive explanatory model of a unique human cognitive ability. In short, the development of grammars over time is a multi-stage evolutionary process involving replicating cognitive systems by means of and within their neurocognitive environment.

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